Module 6 Learning Activities

Integration Testing Performance Security and Observability

Activities 1 and 2 Concept checks

Name _____ Section _____ Date _____

Estimated time 45–60 minutes for all four activities • Total 20 points

Use Module 6 as your reference. Work with fictional situations only. No software, live accounts, calculations, or personal records are needed. Write short answers in the spaces provided.

Activity 1 Match the term 5 points

Write the letter of the correct meaning beside each term. Use each letter once. Each correct match earns 1 point.

Term and answer	Choices
1. Contract testing _____	A. Checking that a system returns to a usable and correct state after a disruption.
2. Authorization _____	B. The elapsed time for a defined operation or journey.
3. Logs _____	C. Checking that communicating applications meet their shared exchange expectations.
4. Latency _____	D. Recorded descriptions of events and outcomes in a component.
5. Recovery testing _____	E. Deciding what an identified user or service is allowed to do.

Activity 2 Select the test type 5 points

Choose from contract testing, end-to-end testing, load testing, security testing, and recovery testing. Use each type once. Each correct answer earns 1 point.

1. A team checks whether the producer's StudentCreated message contains the fields and meanings expected by the LMS consumer. Answer _____

2. A team follows registration through the broker, verifies the LMS account, and checks the required account-ready notice. Answer _____

3. A team observes the integration while it handles expected ordinary registration activity. Answer _____

4. A team checks that a test student cannot read another fictional student's account record. Answer _____

5. After a controlled restore, a team checks that records are usable and pending work resumes without repeated effects. Answer _____

Activity 3 Read a simple failure case

Activity 3 carries 5 points. Read the fictional diagnostic summary, then answer using one or two short sentences. This summary is already sanitized and contains no real student information.

Fictional event summary

Item	Observed information
Event	Event Alpha • StudentCreated
Version	Supported by the consumer
Required student reference	Missing
Broker result	Publication accepted
LMS result	Event rejected; no account created
Diagnostic category	Required student reference is missing

The registrar is an approved publisher. An account-ready notice has not been sent. The team needs a safe way to resolve the rejected event.

Question 1 Identify the problem 2 points

What prevents account creation? Explain why broker acceptance does not mean the account is ready.

_____

_____

_____

Question 2 Choose a safe response 2 points

What should the authorized team correct or check before trying again? State why repeatedly sending the same unchanged event would not help.

_____

_____

_____

Question 3 Protect information 1 point

Name one kind of sensitive information that should be left out of a shared diagnostic screenshot or report.

_____

_____

_____

Reminder: A short, clear explanation is enough. You do not need to write code or create a real message.

Activity 4 A guided test safeguard and alert plan

Activity 4 carries 5 points. Complete the five planning items below. Each suitable response earns 1 point. Use the choices where provided; a short phrase is enough for the final two answers.

Planning situation

The registrar publishes a valid fictional StudentCreated event. The LMS consumer is temporarily unavailable. The broker retains pending work. When the consumer becomes available again, the team wants the account created correctly without repeating a protected business effect.

1 Select the test purpose

Circle one: Resilience and recovery testing / Visual appearance testing

_____

2 Choose the truthful progress state

While the account does not yet exist, circle one: Account provisioning pending / Account ready

_____

3 Select a safeguard during recovery

Circle one: Keep duplicate-safe processing active / Disable input validation

_____

4 Write a simple useful alert

Write a short alert message that would help the operator notice that pending account work is not progressing.

_____

_____

_____

5 Name the first investigation check

What component or dependency should the operator inspect first when the LMS consumer is unavailable?

_____

_____

_____

Use ordinary language. An alert should describe the condition, and the first check should relate to the failed processing boundary.